

Vacuum Operating SOP — Nilfisk CFM 118EXP (AlSi10Mg / Si Powder Pickup)

What this is. The *operating* SOP: how to set up, continuity-verify, run, shut down, clean, and store the Nilfisk CFM 118EXP explosion-proof vacuum when picking up AlSi10Mg or elemental Si powder. It is the companion to the *disposal* SOP ([edison/powder-disposal-sop/powder-disposal-sop.md](#)), which takes over at the point a sealed liner leaves the bin.

Provenance. Consolidated from §13–§17 of [vacuum-accessories-options.md](#) , which are built on three Edison Scientific `LITERATURE_HIGH` reviews — [d7ff65b9](#) (verification of the first-test-run procedure, 20 refs), [86586e0a](#) (crevice-tool continuity anomaly, 10 refs), and [56c08a19](#) (disposal, 15 refs) — plus the **2026-07-06 BYU Risk Management / Waste Management meeting** (Bryant, Steve, Jill, Ed), the **118EXP OEM manual** (UL 1213 par. 22 continuity requirement), the **AlSi10Mg and Si SDS** files in this repo, and Nilfisk tech-support calls/emails through 2026-09-03. Nothing here is new research; it is the same guidance reorganized into the order an operator actually performs it.

Standards basis: NFPA 484, NFPA 652, NFPA 654, NFPA 77, OSHA 29 CFR 1910.157, UL 1213 par. 22 (via the OEM manual).

⚠ AI-synthesized working procedure, not a regulatory ruling. BYU EHS / Risk Management and the faculty PI must review and approve it, and it must be folded into the site-specific NFPA 652 Dust Hazard Analysis (DHA), **before** use with live powder.

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STANDARD OPERATING PROCEDURE: Vacuuming AlSi10Mg / Si Powder with the Nilfisk CFM 118EXP

Facility: BYU Vertical Cloud Lab (VCL)

Equipment: Nilfisk CFM 118EXP explosion-proof vacuum · antistatic main filter (+ clamp) · upstream HEPA · downstream HEPA · conductive hose 107413543 · cuff 4072000343 · aluminum wand 01768601 · crevice tool 01768900 · dust brush 01719401

Applicable powders: AlSi10Mg and elemental Si (aluminum family). **Not** for iron-, copper-, nickel-oxide-bearing, or titanium powders without the alloy-switch procedure in §6.

0. GATES — DO NOT RUN LIVE POWDER UNTIL THESE ARE CLOSED

#	Gate	Status as of 2026-09-04
0.1	Crevice tool 01768900 reads $\leq 0.1 \Omega$ tool-to-wand. The last bench measurement (2026-08-19) was open / MΩ — an anodized socket bore leaves the tool an <i>isolated conductor</i> in the powder stream. The OEM manual's own rule applies: " <i>if unacceptable results are recorded, DO NOT OPERATE THE CLEANER.</i> "	 Open. Tool is out of service. Run the §1.4 remediation ladder, or run the first test with the dust brush alone (0.03 Ω , passes).
0.2	Class D extinguisher (Met-L-X / Lith-X / NaCl-based) staged at the station — ≤ 75 ft travel distance per OSHA 1910.157, ideally ≤ 25 ft. Never water, CO ₂ , or ABC on this powder.	 Not confirmed procured.
0.3	ESD wrist strap decision. A strict reading of NFPA 484 §15.3.1.2 calls for the operator to be grounded (sub-micron fines can have MIE < 30 mJ). This was not on the 2026-07-06 Risk-Management PPE list, and the epoxy/vinyl floor rules out ESD-footwear grounding.	 Reconcile with EHS — then edit §2 accordingly.
0.4	Nilfisk clearance in writing. The 2026-09-03 phone call ("probably anodized, good to use if everything else is low") conflicts with the 2026-08-31 written answer ("the crevice nozzle should be measuring close to 0 Ω , not in M Ω "). Get the resolution by email, referencing UL 1213 par. 22.	 Requested.
0.5	EHS + PI sign-off on this operating SOP (not only the disposal SOP) and a documented NFPA 652 DHA on file.	 Pending.

1. PRE-USE CONTINUITY VERIFICATION (REQUIRED BEFORE EVERY USE)

The OEM manual is explicit: "*THE OPERATOR SHOULD TEST THE VACUUM MACHINE FOR GROUND CONTINUITY BEFORE EACH USE,*" checked "*between each connection ... tool to wand, wand to hose, hose to machine,*" and the path "**shall not exceed 0.1 ohm**" (UL 1213 par. 22). This is an OEM requirement, not merely an NFPA-derived best practice.

1.1 Acceptance criteria

Path	Criterion	Notes
Each metal-to-metal connection — tool → wand, wand → coupler, coupler → hose cuff, hose → machine inlet	$\leq 0.1 \Omega$	The governing gate. Any single failing joint blocks the run.

Path	Criterion	Notes
Assembled path — tool tip → wand → coupler → hose → vacuum inlet	$\leq 0.1 \Omega$ per connection; hose contributes its own resistance	Measure assembled, as the manual intends.
Hose end-to-end	$\sim 4 \Omega$ typical; spec $R \leq 10^4 \Omega$	Nilfisk-confirmed normal for their conductive hose. The hose is a listed component , not a "connection," so it is judged against its own spec — it passes.
Vacuum body / bin → building ground	$< 1 \Omega$	Body sections measured $\sim 0.2 \Omega$, lid-to-ground-cable $\sim 1 \Omega$.
Conductive liner (seated) → bin / ground	$< 1 \text{ k}\Omega$	Re-check after every liner change. There is no bonding clip on this unit — grounding is by liner-to-bare-bin contact.
Dust brush ferrule → wand	$< 10 \Omega$ (measured 0.03Ω )	Also confirm the bristles are the conductive EXP variant — a bonded ferrule with nylon bristles is still a floating path at the tips.
Any metal object in the work zone → ground	$< 10 \Omega$	Isolated-conductor sweep, §2.3.

1.2 Meter and technique

- Use a milliohm-capable meter (the EXTECH 380560 used for the baseline resolves 0.01Ω); a general-purpose DMM cannot resolve 0.1Ω meaningfully. Zero/REL the leads first.
- **Probe the marked points.** Standardized probe points are marked on the vacuum and tools; use them every time so readings are comparable run-to-run.
- Press **firmly**, hold **still**, and maximize contact area (side of the probe, not the point). Sloppy technique swings readings by orders of magnitude; good technique repeats to $\sim 10 \text{ m}\Omega$.
- **Record every reading in the session log** (§9) and compare against the previous run — a drifting joint is the early warning.

1.3 If everything passes

Proceed to §2. If anything fails, **stop** — do not run powder.

1.4 If a joint fails — remediation ladder (do them in this order)

Applies to the crevice tool (gate 0.1) and to any joint that goes open. **Do not abrade any surface you can see** — an anodized skin *over grounded metal* is harmless (it breaks down at $\sim 20\text{--}125 \text{ V}$, far below the $\sim 4 \text{ kV}$ propagating-brush-discharge threshold). The only problem is a tool body that **floats**, so the only place bright metal is needed is inside the socket bore.

1. **Probe the anodizing rack marks** — the small bare spots where the part hung during anodizing, often inside the bore or at sharp edges. You may get continuity with **zero** abrasion.
2. **Seat the tool fully and tightly** on the wand and re-measure — an interference fit can scrape through the skin on its own. **No abrasion.**
3. **Plastic-safe contact cleaner** (e.g. CRC QD) on the bore and wand tip, re-seat, re-measure. Clears oil/oxide films; will not touch anodize. **No abrasion.**

4. **Only if 1–3 fail:** hand-abrade a **5–10 mm contact band inside the socket bore** and the mating band of the wand tip with **gray or maroon Scotch-Brite** (not green). **Hand only — no Dremel, no grinder** (power tools generate far more combustible Al fines and gouge the seat). Do this **before** the tool has touched powder, away from ignition sources, and capture debris with an IPA-dampened wipe.
5. **Re-verify $\leq 0.1 \Omega$ assembled, record the value**, and document the remediation in the DHA — the manual treats bore work as an "alteration."
6. **If it still won't reach $\leq 0.1 \Omega$, stop and replace the tool.** Do not escalate the abrasion.

2. SETUP — BEFORE THE POWDER COMES OUT

2.1 PPE (all steps below)

- **N95 or P100 respirator** — the AlSi10Mg SDS lists **H334** (respiratory sensitizer).
- **Nitrile gloves** (non-sparking; no latex).
- **Cotton or antistatic lab coat.** The black VCL coat is 100% cotton ✓; the atomizer coat is antistatic ✓; the 80/20 poly-cotton white coat is **not acceptable** ✗ — it builds static.
- **Safety glasses.**
- Closed-toe shoes.
- **ESD wrist strap** — pending the gate-0.3 decision with EHS; wear one if available.

2.2 Machine and station

- ☐ **\$1 continuity check passed** and readings logged.
- ☐ **All three filters installed** — antistatic main filter (with clamp), upstream HEPA, downstream HEPA.
- ☐ **Conductive liner seated** in the bin, in contact with bare bin metal, lining the interior; liner-to-bin **$< 1 \text{ k}\Omega$** verified. Black carbon-loaded PE only — never pink "antistatic" or silver shield bags.
- ☐ **Class D extinguisher** staged and within reach (gate 0.2).
- ☐ **Grounded steel interim pail** staged with its lid available, labeled per the disposal SOP.
- ☐ **Nothing wet, hot, or oxidizing** on the bench; no water, no solvents at the station, no compressed air anywhere near this task.

2.3 Work zone — isolated-conductor sweep (the most important addition)

The hazard is **ungrounded metal**, not an unbonded floor. An insulating floor (concrete, epoxy, vinyl) cannot act as the capacitor that produces a spark, and NFPA 484 §15.3.1.1 requires *the equipment* to be grounded and bonded — not the surface being cleaned. Vacuuming powder off a floor or off the atomizer housing **is the job**, and the bonded wand → coupler → hose → vacuum chain is the control that makes it safe.

- ☐ Walk the area and **bond or remove every ungrounded conductor** near the powder: metal tray, cart, step stool, loose sheet-metal panel, metal trash can, unattached duct stub. Anything metal that stays gets a bonding wire to the vacuum body or the same building ground point, **verified $< 10 \Omega$** .
- ☐ **Confirm the atomizer housing / frame is bonded to building ground ($< 10 \Omega$)**. This is the full-scale version of the same rule — a large metal enclosure in direct contact with charging powder — and it matters far more than what the powder is sitting on.

- ☐ **Area control:** restrict the immediate area to trained, PPE-equipped personnel; post signage — "COMBUSTIBLE METAL POWDER IN USE — Authorized Personnel Only — NO Water, NO Compressed Air."
- ☐ Second person available (and recording video, for the training record).

2.4 For the first test run specifically

- ☐ Use a **few grams** of AlSi10Mg over a **small, defined footprint you can verify clean** afterward against the NFPA 654 **1/32-inch layer-depth** criterion — not spread widely or where fines can travel under equipment.
- ☐ The floor or housing surface is acceptable and is representative of the real job; a **bonded** metal tray at bench height gives better nozzle control if the run allows it (ergonomics preference, not a safety gate).
- ☐ Run end-to-end — pickup, suction, sealing, disposal handoff — before scaling up.

3. VACUUMING

Goal: keep powder out of the air. Sub-micron fines can have MIE < 5 mJ; bulk AlSi10Mg is 80–350 mJ.

1. **Equalization touch.** Before the wand tip approaches the powder, briefly touch the grounded wand tip to **any grounded metal point** — the vacuum body, a bonded frame, the bench ground — *away from the powder*.
2. **Vacuum ON before** you approach the powder.
3. **Move slowly.** Slow, deliberate passes; let the airflow do the work. Fast motion, dropping, and piling are what put powder airborne.
4. **Keep the nozzle close to the surface** so capture velocity is highest right at the powder.
5. **Never blow powder toward the nozzle**, and never use compressed air anywhere near this task (NFPA 484 prohibition; AlSi10Mg SDS §8.2).
6. **Lift the tool away from the powder before switching OFF**, so no loose pile is left at the tip.

Crevice nozzle (⚠ gated by 0.1) — corners, seams, gasket grooves, around fittings, small spills. Its small opening gives the highest tip velocity, which is what dense settled metal powder needs. Draw it steadily along the crevice; don't stab at piles.

Dust brush — adherent fines on flat/contoured surfaces (μm powder clings electrostatically and won't lift with suction alone). Use **brush-and-capture**: let the bristles just kiss the surface to release powder while the airflow at the ferrule takes it — **don't sweep powder across the surface into the air**.

Watch continuously for:

- **Visible airborne dust** → slow down, get the nozzle closer.
- **Bin fill** → never exceed ~25%.
- **Warmth or odor** → stop. Never vacuum hot or glowing particles; let any powder off a laser/furnace/hot plate cool to room temperature first.

4. SHUTDOWN — IMMEDIATELY AFTER VACUUMING (EVERY SESSION)

1. **Clear the bores:** with the tool in **clean air** (not sitting in powder), run the vacuum **15–30 s** to pull residual fines out of the hose/wand bores into the liner.
2. **Power off. Wait ≥ 60 s** for airborne powder to settle **before opening or disassembling anything**.

3. **Check liner fill.** If > ~25%, seal and remove now (step 4). If below and you are not switching powders, it may stay for the next session.
4. **Seal and transfer** (this is the handoff to the disposal SOP):
 - Gently gather the liner above the powder, **gooseneck-twist 2–3 turns**, secure with **conductive/ESD tape or two steel zip ties**.
 - Lift the sealed liner into the **grounded steel interim pail**.
 - **Set the lid on — do NOT crimp it.** (Waste Management: crimping raises the explosion hazard; the un-crimped lid vents trace H₂.)
 - Install a **fresh conductive liner**, seated against bare bin metal; re-verify liner-to-bin < **1 kΩ**.
5. **Clean the tools** (§5) and **store them** (§6).
6. **Log the session** (§9). Contact BYU Waste Management for pickup when the pail warrants it; the pail is on a **1-year clock** from the date EHS tags it.

5. CLEANING THE TOOLS AFTER USE

5.1 Same alloy family (AlSi10Mg ↔ Si ↔ other Al alloys) — routine care

A full IPA cleaning is **not** required every time.

1. **Dry-wipe FIRST.** Remove all visible powder with a dry lint-free wipe (Kimwipes are acceptable).
2. **Then a light IPA wipe — dampened, not wet. Never apply IPA to a visibly powder-coated surface:** IPA vapor over combustible metal dust forms a **hybrid mixture** with a *lower* ignition energy than either alone. The AlSi10Mg SDS also lists **alcohols** among incompatible materials — so: sparing, dry-wipe-first, never a soak.
3. Prefer **70% IPA** (higher flash point than 99%); work with ventilation on and ignition sources away.
4. **Let it evaporate fully — ≥ 5 min, until there is no IPA smell** — before reassembly, bagging, or any contact with powder.
5. **Dust brush: do not wet the bristles.** Tap them out gently into the sealed liner with the vacuum running, then wipe the **ferrule/body only**. Soaked bristles hold both solvent and fines and dry slowly.
6. **Re-verify continuity after any cleaning or reassembly** (§1).

Never: water, acetone, or chlorinated solvents (the SDS notes a violent reaction with halogenated hydrocarbons); never blow out, wash, or rinse the filters or attachments.

5.2 How to wet the wipe — do not decant IPA into a spray bottle

Spraying is routine lab practice elsewhere; it is specifically wrong here. Atomizing a flammable liquid over combustible metal fines is exactly the hybrid-mixture case (a mist ignites below the bulk flash point); the plastic bottle and the charged droplets reintroduce the charged insulator this whole setup exists to eliminate; overspray lands on powder; the jet disperses powder like the prohibited compressed air; an N95 gives **no** protection against solvent vapor; and a decanted bottle is a GHS-labeled secondary container that must not live at the vacuum station.

Approved alternatives, in order:

Option	Why it fits
Pre-saturated 70% IPA wipes (best)	Exactly damp-not-wet, no bulk solvent at the bench, no aerosol, no dispensing static. E.g. Berkshire SatPax 1000 (70% IPA, 6×9", 100/canister) or Texwipe TechniSat TX1045.
Steel plunger dispensing can	Press the wipe on the plunger for a metered wet spot — no aerosol; the can is metal, bondable, and FM/UL-listed with a flame-arrester pan (e.g. Justrite 1-pt/1-qt).
Pour-and-dab, bottle closed	Free. At a ventilated spot away from the powder , tip a small amount onto the wipe, re-cap immediately , carry the damp wipe to the tool.

Keep bulk IPA in the flammables cabinet — **not** at the vacuum station.

5.3 Where the used wipes go

A Kimwipe loaded with AlSi10Mg **is** combustible metal dust — it does not go in the regular trash.

- **Dry wipes** → into the vacuum's conductive liner (or a small conductive bag that goes into it) with the collected powder.
- **IPA-damp wipes** → lay flat in a ventilated spot until the IPA has **fully flashed off** (≥ 5 min, **no smell**), *then* bag them with the dry wipes. **Never** drop a solvent-wet wipe into the sealed liner or the interim pail — the un-crimped lid is for trace H₂ venting, not solvent vapor.
- Confirm this handling in one line with Jill/Ed at Waste Management, since the pail is on their tracking system.

6. SWITCHING POWDERS

Situation	What is required
AlSi10Mg ↔ Si ↔ other Al-family	Compatible. Liner swap + dry wipe + visual check. Same tools, same pail.
Any iron-, copper-, or nickel-oxide-bearing powder, or Ti	Incompatible — thermite precursors. Replace the main filter and both HEPA filters (dispose as contaminated waste), full IPA clean of bin/hose/wand/tools, fresh liner, fresh dedicated interim pail , and re-verify continuity before use.

Keep this hose/wand/crevice/brush set **dedicated and labeled** as the aluminum-family set. The dust brush especially retains powder in its bristles — never share it across families.

7. STORING THE WAND AND ATTACHMENTS

Powder left inside a wand, crevice nozzle, or brush **is** an accumulation of combustible metal dust.

- **Clean first, then store** (\$5). Don't store visibly powder-coated tools.
- **Store dry** — no moisture near stored tools (damp Al fines can self-heat). Away from sinks, eyewash, and water lines.
- **Contain residual dust:** cap/plug the wand and crevice-tool openings, or slip each tool into a **black carbon-loaded conductive PE bag** (same liner stock, small). One shared bag for same-family tools is fine — label it.
- **Segregate** from oxidizers, metal-oxide powders, and ignition sources — store with/near the grounded vacuum station, not in a drawer of random chemicals.

- Keep the **storage area** itself clean to the NFPA 654 **1/32-inch** layer-depth criterion.
- Don't let sealed powder-loaded tools and full liners accumulate — collected powder belongs in the liner → pail → EHS stream.

If a tool truly can't be cleaned before storage (end of a long session), treat it as containing combustible dust: cap/bag it, keep it dry and grounded, and clean it before the next use.

8. SPILL RESPONSE

- **NEVER sweep** — sweeping creates the dust cloud.
- Scoop visible piles with a **non-sparking (plastic or brass) scoop** into a conductive bag.
- Re-vacuum only after confirming the ground chain is intact.
- For any visible layer on the floor: restrict access, ventilate, and call EHS.
- Fire: **Class D only**. Never water, CO₂, or ABC.

9. ABSOLUTE PROHIBITIONS

Never	Why
Water, or wet cleaning of anything powder-contaminated	$\text{Al} + \text{H}_2\text{O} \rightarrow \text{H}_2$ (SDS H261); Class D fire risk
Compressed air to blow powder or clean tools/filters	Creates an explosive dust cloud (SDS §8.2; NFPA 484)
Acetone or chlorinated solvents	Violent reaction with halogenated hydrocarbons
Spraying/atomizing IPA near the powder	Hybrid mixture; mist ignites below bulk flash point (§5.2)
IPA on visible powder, in the liner, or on filters	Hybrid mixture; lower MIE than either alone
Running with any joint failing the $\leq 0.1 \Omega$ check	OEM: "DO NOT OPERATE THE CLEANER"
Ungrounded metal objects in the work zone	Isolated conductor: ~5–10 mJ at ~10 kV vs. fines MIE < 5 mJ
Crimping the interim-pail lid	Blocks trace H ₂ venting; raises explosion hazard
Vacuuming hot or glowing particles	Direct ignition source
Mineral-oil (or any) passivation of the collected powder	Adds a Class IB flammable liquid; dry sealed grounded containment is sufficient at this scale
The 80/20 poly-cotton white lab coat	Builds static
Bin above ~25% fill, or pail above ~50%	Accumulation limits set for this SOP

10. SESSION LOG (record every run)

Date • operator • powder type • approximate mass vacuumed • **all §1 continuity readings** • liner and pail fill after the run • any anomaly (airborne dust, warmth, odor, suction loss) • video reference. Drifting continuity readings are the early warning of a failing bond — log the numbers, don't just check a box.

OPERATOR QUICK-REFERENCE CARD — 118EXP POWDER PICKUP

PPE: N95 · nitrile gloves · cotton/antistatic lab coat (not the white poly one) · safety glasses · ESD strap if issued.

Before

- ☐ Continuity: every joint $\leq 0.1 \Omega$ · hose $\sim 4 \Omega$ · body-to-ground $< 1 \Omega$ · liner-to-bin $< 1 \text{ k}\Omega$ — log the numbers. *Fail = do not run.*
- ☐ Crevice tool cleared ($\leq 0.1 \Omega$) — otherwise **dust brush only**.
- ☐ 3 filters in (main + clamp, upstream HEPA, downstream HEPA).
- ☐ Fresh conductive liner seated against bare bin metal.
- ☐ Class D extinguisher staged · interim pail + lid staged.
- ☐ Bond or remove all loose metal in the zone ($< 10 \Omega$), including the atomizer housing.
- ☐ Signage up · second person present/recording · small defined footprint of powder.

During

- ☐ Touch wand tip to grounded metal (away from powder) → **vacuum ON** → approach.
- ☐ Slow passes · nozzle close · brush just kisses the surface · never blow powder.
- ☐ Watch: airborne dust · bin $\leq 25\%$ · any warmth/odor → **stop**.
- ☐ Lift tool clear of the powder **before** switching off.

After

- ☐ 15–30 s run-on in clean air → **power OFF** → **wait ≥ 60 s**.
- ☐ Liner $> 25\%$? → gooseneck twist $\times 2-3$, ESD tape / 2 steel ties → into the pail → **lid on, NOT crimped** → fresh liner.
- ☐ Dry-wipe tools first, **then** a barely-damp 70% IPA wipe (never spray; brush bristles stay dry) → let dry ≥ 5 min / no smell.
- ☐ Dry wipes into the liner; IPA wipes flash off first.
- ☐ Cap/bag tools, store dry with the aluminum-family set.
- ☐ Log the session.

Emergency: Class D extinguisher only — **no water, no CO₂, no ABC**. Evacuate, alert others, call BYU Risk Management. Never sweep a spill.